



# Severity-aware coordination of grid-forming and grid-following BESS for stability enhancement in weak transmission networks

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## ABSTRACT

The displacement of synchronous machines with inverter-based renewable sources has intensified concerns over frequency instability, voltage dips, and weak fault resilience in transmission networks. While Battery Energy Storage Systems (BESS) are increasingly deployed for stability support, limited research has addressed coordinated operation of grid-following (GFL) and grid-forming (GFM) inverters under renewable-dominated conditions. This paper proposes a severity-aware BESS control framework that blends GFL and GFM contributions through a transition mechanism governed by a severity index, ensuring efficient current-following under mild disturbances and robust voltage-forming support during severe faults. A systematic parametric analysis of GFM:GFL penetration ratios is conducted, where frequency and voltage stability metrics are consolidated into a composite index and optimised to quantify system performance. Results demonstrate monotonic improvements in frequency nadir, rate-of-change-of-frequency (RoCoF) suppression, and voltage resilience with increasing GFM share. Furthermore, a comparative evaluation of Droop and Virtual Synchronous Machine (VSM) strategies at intermediate GFM shares highlights Droop's faster transients against VSM's superior damping and black start capability. Collectively, the findings confirm that severity-aware GFM-GFL coordination offers a practical pathway to resilient operation of renewable-rich weak grids, directly advancing the objectives of Sustainable Development Goal (SDG) 7 on Affordable and Clean Energy.

## 1. Introduction

The rapid transition toward renewable-dominated power systems is transforming conventional grid dynamics [1]. The displacement of synchronous generators (SGs) with inverter-based renewable energy sources (RESs), such as solar photovoltaics and wind turbines, has resulted in declining system inertia and increased vulnerability to frequency and voltage instability, especially in weak transmission networks characterized by low short-circuit ratios (SCR), poor damping, and high rates of change of frequency (RoCoF) [2,3]. In this context, Battery Energy Storage Systems (BESS) have emerged as key stabilizing assets due to their rapid response capabilities and controllability [3,4]. However, the effectiveness of BESS depends heavily on the control strategy employed at the inverter interface, particularly under low-inertia, fault-prone grid conditions [1,2].

Two dominant control approaches are employed for grid-interfaced inverters: Grid-Following (GFL) and Grid-Forming (GFM). GFL inverters synchronize to the grid using a phase-locked loop (PLL) and inject current based on a reference, making them susceptible to instability in weak

grid environments and incapable of black start operations [5–7]. GFM inverters, in contrast, regulate voltage and frequency independently, emulating synchronous machine behavior and offering advantages such as inertia provision, frequency support, and fault ride-through (FRT) [8–10]. Various GFM control strategies have been developed, including droop control [10], virtual synchronous generator (VSG), compensated generalized VSG (CGVSG), adaptive VSG (AVSG) [11], virtual oscillator control (VOC) [12], and most recently, frequency-following voltage-forming (FFL-VFM) schemes [13].

While these strategies have shown promise individually, the literature still lacks a cohesive framework for integrating and coordinating GFM and GFL inverters within a unified BESS control architecture. Previous studies have shown that significant performance gaps exist between GFM and GFL modes, particularly under fault and islanding scenarios [10,14]. Advanced control techniques, such as fictive dq control, have been proposed to manage unbalanced load conditions in mini-grid environments using GFM-based BESS, but these approaches remain limited in scope [6]. Dynamic modelling efforts have successfully integrated GFM inverter behaviour into industry-standard tools like Power System

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Simulation for Engineering (PSS®E) and MODES, yet coordination with GFL counterparts has not been systematically addressed [8]. Several works have carried out comparative analyses of GFM strategies, such as droop control, VSG, CGVSG, and AVSG based on time-domain and impedance-domain metrics; however, these studies typically examine each strategy in isolation rather than within mixed GFM-GFL deployments [11]. Moreover, while the influence of inverter penetration levels on frequency response and damping has been analytically explored [10], and the impact of BESS control on frequency containment under high renewable penetration has been evaluated using large-scale simulation frameworks [9], coordinated control between inverter types and the effect of varying GFM:GFL ratios have remained largely unexamined. Additional work has replicated real-world blackout events to validate the FRT benefits of GFM inverters, yet still did not consider multi-inverter coordination strategies [15].

Recent research has increasingly focused on addressing the remaining gaps in coordinated inverter control. Comparative benchmarks of GFM and GFL BESS under real-world load-shedding events have revealed notable differences in RoCoF and frequency nadir across multiple operational modes [16,17]. Comprehensive taxonomies have also been developed to classify existing GFM control strategies while outlining future research directions, including hybrid angle-control schemes and Multiple-Input Multiple-Output (MIMO)-based architectures [12]. To improve frequency stability assessments under realistic system conditions, new stability indices have been proposed that incorporate stochastic variations in renewable generation and load demand [18]. Advances in short-circuit modeling of inverter-interfaced generation stations (IIGS) under low short-circuit ratio (SCR) conditions have further introduced decoupled sequence-control approaches, improving both convergence and modeling accuracy [19]. Control-design developments have explored the decoupling of frequency-following and voltage-forming behaviors in inverters, leading to hybrid architectures particularly well suited for constant-power renewable sources [13,20]. Moreover, coordinated control frameworks based on leader-follower consensus mechanisms have demonstrated enhanced multi-inverter operation, enabling improved frequency regulation, reactive-power sharing, and resilience to communication disturbances in microgrids [21].

Despite these advances, several important research challenges remain unaddressed. A key limitation is the absence of a unified control framework that dynamically coordinates GFM and GFL inverter modes within BESS, particularly during fault and disturbance scenarios in weak grids, where most existing studies continue to treat these control strategies in isolation [5,11,21,22]. Additionally, there is a noticeable lack of systematic investigations into how varying the sharing of GFM and GFL inverters, affects critical stability indicators like RoCoF, frequency nadir, and FRT performance in low-inertia systems [15,16,23,24]. Furthermore, although a range of GFM strategies have been proposed in the literature, comparative evaluations under consistent test conditions remain scarce. In particular, the relative strengths and weaknesses of droop control and VSG approaches are rarely assessed side by side within standardised weak grid environments, making it difficult to draw meaningful conclusions about their suitability for next-generation inverter-based systems [11].

Building on these observations, this work addresses three major gaps. First, existing studies lack a coordinated GFM-GFL architecture capable of adapting contributions based on real-time disturbance severity. Second, no prior work offers an optimisation-based analysis of GFM:GFL penetration using unified frequency and voltage metrics. Third, fair comparisons of GFM controls, such as Droop and VSM, under identical weak-grid conditions remain absent. This study fills these gaps by proposing a severity-aware blended GFM-GFL coordination scheme, an optimisation-driven penetration analysis, and a unified evaluation of Droop and VSM controls, including black-start capability.

Table 1 further highlights these gaps, indicating that prior studies rarely integrate GFM-GFL coordination, penetration-level assessment,

and weak-grid validation within a unified analytical framework. The present work is the first to provide such a comprehensive integration.

The workflow is summarized in the overall methodology framework shown in Fig. 1. The main contributions of this work are as follows:

- **Severity-Aware BESS Control Framework:** A novel severity-aware BESS coordination framework is proposed in which GFL and GFM modes operate cooperatively, guided by a severity index and validated against grid-code fault-clearing times. Unlike conventional approaches that treat the two modes as independent or mutually exclusive, the framework enables GFL to deliver efficient current-following during mild disturbances while GFM progressively assumes voltage-forming dominance under prolonged or severe faults. This adaptive interaction improves resilience in weak grids while preserving efficient operation under normal conditions.
- **Parametric Analysis of GFM and GFL Shares:** Building on the coordination framework in Contribution 1, a systematic parametric analysis of GFM-GFL shares is conducted using normalised frequency metrics and voltage stability margins within a unified composite stability measure. Unlike prior studies that assess inverter penetration only qualitatively or under isolated scenarios, this analysis provides a structured, quantitative evaluation of how varying GFM:GFL ratios affect weak-grid dynamics. Results show consistent monotonic gains in frequency and voltage resilience as GFM share increases, offering evidence-based insight into the stabilising role of GFM in weak transmission networks.
- **Comparative Evaluation of Droop and VSM Control:** Extending the previous parametric analysis, a unified comparison of two GFM control strategies, Droop and VSM, is performed at representative GFM penetration levels of 50% and 75%. Unlike earlier studies conducted under inconsistent network assumptions, this evaluation is carried out within a coordinated GFM-GFL framework, ensuring fair and meaningful benchmarking. The results highlight clear behavioural differences: Droop provides faster primary response, while VSM offers stronger damping and black-start capability, giving practical guidance on their suitability for future high-IBR grids.

Collectively, these contributions advance robust BESS control strategies for inverter-dominated renewable-rich systems, providing a foundation for resilient operation in weak grids. The paper is organised as follows: Section 2 outlines the proposed methodology, Section 3 describes the network and cases, Section 4 discusses results, and Section 5 concludes with key findings and future work.

## 2. Proposed methodology

This section outlines the proposed BESS control methodology, comprising: (i) a severity-aware GFL-GFM blending scheme, (ii) multi-objective optimisation of GFM:GFL penetration, and (iii) comparative assessment of Droop and VSM at mid-level penetration. The overall framework appears in Fig. 1.

### 2.1. Severity-aware coordination framework

The first contribution introduces a severity-aware BESS control framework in which GFM and GFL operate in *coordination* during grid disturbances, rather than in an exclusive switching manner. The aim is to exploit the current-following efficiency of GFL under mild conditions while simultaneously enabling voltage-forming robustness of GFM during severe faults.

#### 2.1.1. Severity index

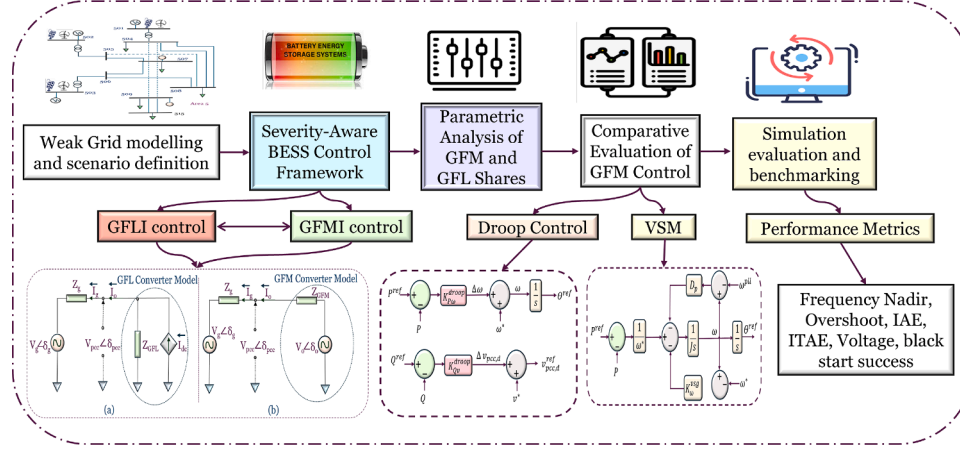
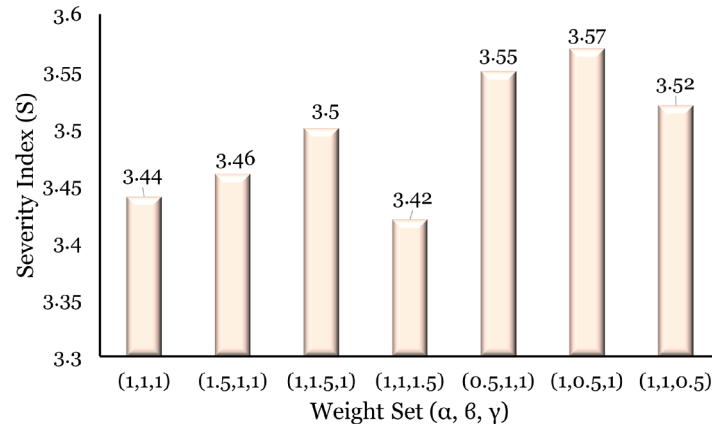
A dimensionless severity index  $S(t)$  is defined to quantify the disturbance level, normalised against grid code thresholds for frequency deviation, RoCoF, and voltage deviation:

$$S(t) = \alpha \frac{|f(t) - f_{\text{ref}}|}{\Delta f_{\text{max}}} + \beta \frac{\left| \frac{df}{dt} \right|}{\text{RoCoF}_{\text{max}}} + \gamma \frac{|V_{\text{ref}} - V(t)|}{\Delta V_{\text{max}}} \quad (1)$$

**Table 1**

Comparison of studies on GFM-GFL coordination and weak-grid, renewable-dominated system performance.

| Ref       | Severity-Aware GFM-GFL Co-ordination | Comparative Evaluation of GFM Control Strategies | Parametric Analysis of GFM:GFL Ratios | Assessment under Weak Grid Conditions | Renewable-Dominated System Context |
|-----------|--------------------------------------|--------------------------------------------------|---------------------------------------|---------------------------------------|------------------------------------|
| [5]       | ×                                    | ×                                                | ×                                     | ✓                                     | ✓                                  |
| [6]       | ×                                    | ×                                                | ×                                     | ✓                                     | ✓                                  |
| [9]       | ×                                    | ×                                                | ✓                                     | ✓                                     | ✓                                  |
| [11]      | ×                                    | ✓                                                | ×                                     | ✓                                     | ✓                                  |
| [13]      | ×                                    | ×                                                | ✓                                     | ✓                                     | ✓                                  |
| [16]      | ×                                    | ×                                                | ×                                     | ✓                                     | ✓                                  |
| [21]      | ×                                    | ×                                                | ×                                     | ✓                                     | ✓                                  |
| This Work | ✓                                    | ✓                                                | ✓                                     | ✓                                     | ✓                                  |

**Fig. 1.** Overall proposed methodology framework.**Fig. 2.** Sensitivity of S to variations in the weighting coefficients  $\alpha$ ,  $\beta$ , and  $\gamma$ .

where  $\Delta f_{\max}$ ,  $\text{RoCoF}_{\max}$ , and  $\Delta V_{\max}$  denote the allowable limits for frequency deviation, RoCoF, and voltage deviation. The severity index applies grid-code normalisation to keep all metrics dimensionless and comparable, enabling equal weights ( $\alpha = \beta = \gamma = 1$ ) as a neutral baseline. Sensitivity tests ( $\pm 50\%$ ) show only minor variation ( $\approx 3.4$ – $3.6$ ; Fig. 2), indicating that the GFM-GFL coordination remains robust to weighting choices.

### 2.1.2. Fault duration and grid-code validation

To validate the coordination mechanism, disturbances are modelled at distribution-level buses (20 kV). According to the National Electricity Rules (NER) [25], systems at 100 kV and below follow “as necessary” clearing guidelines. In this study, two representative cases are applied, namely a fault duration of  $t_{\text{fault}}^{(501)} = 60$  ms at bus 501 (20 kV) and  $t_{\text{fault}}^{(502)} = 100$  ms at bus 502 (20 kV). The first represents a disturbance

cleared within the fast-protection time frame, while the second represents an extended disturbance within the backup protection range. This setup aligns with NER practices while enabling validation of the proposed control.

### 2.1.3. Coordinated mode transition logic

Instead of exclusive mode switching, a *blended control law* governs inverter operation during disturbances. The coordinated reference is expressed as:

$$u^*(t) = (1 - \lambda) u_{\text{GFL}}^*(t) + \lambda u_{\text{GFM}}^*(t); \lambda(t) = \min \left( 1, \frac{S(t)}{S_{\text{th,high}}} \cdot \frac{t_{\text{fault}}}{t_{\text{clear,max}}} \right) \quad (2)$$

where  $\lambda(t)$  is the hybrid blending factor in  $[0, 1]$  that sets the relative GFL-GFM contribution. It increases with the severity index  $S(t)$  and the ratio  $t_{\text{fault}}/t_{\text{clear,max}}$ . Thus,  $\lambda = 0$  gives pure GFL,  $\lambda = 1$  gives pure

GFM, and intermediate values yield a smooth hybrid response. For short disturbances ( $\approx 60$  ms),  $\lambda \approx 0.5$  indicates balanced coordination, while more severe faults ( $\approx 100$  ms) drive  $\lambda \rightarrow 1$ , shifting toward dominant GFM support.

#### 2.1.4. Control laws

The coordinated framework employs complementary GFL and GFM control laws that operate simultaneously and transition smoothly based on disturbance severity. In GFL mode, the inverter behaves as a current-controlled device, tracking  $dq$ -axis current references derived from the active and reactive power setpoints:

$$i_{dq}^*(t) = \frac{P^*}{V_{dq}(t)} + j \frac{Q^*}{V_{dq}(t)}, \quad (3)$$

where  $P^*$ ,  $Q^*$  are the commanded active and reactive powers and  $V_{dq}(t)$  is the measured  $dq$ -frame voltage. This mode provides fast current-limited response and effective damping during mild disturbances where PLL synchronisation remains reliable.

In contrast, the GFM mode regulates voltage and frequency directly through droop control:

$$\begin{aligned} \dot{\delta}(t) &= \omega(t) = \omega_{\text{ref}} - K_f (P(t) - P^*), \\ V_{dq}^*(t) &= V_{\text{ref}} - K_v (Q(t) - Q^*), \end{aligned} \quad (4)$$

where  $P(t)$  and  $Q(t)$  are instantaneous powers,  $\delta(t)$  is the phase angle with angular frequency  $\omega(t)$ ,  $\omega_{\text{ref}}$  and  $V_{\text{ref}}$  are nominal settings, and  $K_f$ ,  $K_v$  are the active- and reactive-power droop gains. This mode provides voltage formation, inertia-like support, and improved robustness under severe faults.

Blending this behaviour through the severity-dependent factor  $\lambda(t)$  allows fast GFL response during mild events and a smooth shift to GFM dominance as severity increases, ensuring stable and adaptive performance across varying disturbances.

### 2.2. Multi-objective optimization of GFM:GFL penetration ratios

Building upon the coordination framework, the second contribution evaluates the dynamic impact of varying GFM:GFL shares. A total generation of 330 MW is maintained, with penetration ratios ranging from fully GFL ( $x = 0$ ) to fully GFM ( $x = 1$ ) across bus 502, as shown in Table 2.

**Table 2**  
Allocation of GFM and GFL power across scenarios.

| GFM Share ( $x$ ) | GFM (MW) | GFL (MW) |
|-------------------|----------|----------|
| 0.00              | 0        | 330      |
| 0.25              | 82.5     | 247.5    |
| 0.50              | 165      | 165      |
| 0.75              | 247.5    | 82.5     |
| 1.00              | 330      | 0        |

#### 2.2.1. Stability metrics

To assess dynamic performance under different GFM:GFL ratios, both frequency- and voltage-domain indicators are used. Frequency deviations are measured using the integral of absolute error (IAE) and the integral of time-weighted absolute error (ITAE), defined as:

$$\text{IAE} = \int_0^T |f(t) - f_{\text{ref}}| dt, \quad \text{ITAE} = \int_0^T t \cdot |f(t) - f_{\text{ref}}| dt; \quad (5)$$

In the voltage domain, performance is assessed through the minimum and maximum values of the voltage trajectory, together with their deviation, expressed as:

$$V_{\min} = \min_{t \in [0, T]} V(t), \quad V_{\text{over}} = \max_{t \in [0, T]} V(t), \quad \Delta V = V_{\text{over}} - V_{\min}. \quad (6)$$

where  $V_{\min}$  and  $V_{\text{over}}$  are the minimum and maximum voltages, with  $\Delta V$  their deviation, capturing dips, overshoot, and recovery for the composite stability index.

#### 2.2.2. Composite stability index

The optimization problem is formulated to determine the optimal GFM share  $x \in [0, 1]$  that maximises overall stability performance. The composite stability index consolidates frequency and voltage domain metrics as follows:

$$\begin{aligned} \text{Maximise } J(x) &= w_f \cdot (\alpha_1 m_{f_{\text{nadir}}} + \alpha_2 m_{f_{\text{peak}}} + \alpha_3 m_{\text{IAE}} + \alpha_4 m_{\text{ITAE}} + \alpha_5 m_{T_s}) \\ &+ w_v \cdot (\beta_1 m_{V_{\min}} + \beta_2 m_{V_{\text{over}}} + \beta_3 m_{\Delta V}), \end{aligned} \quad (7)$$

$$m_i = \frac{\text{Margin Value}}{\text{Threshold}}, \quad m_i \in [0, 1]. \quad (8)$$

where  $m$  is the normalised stability margin for each metric relative to its limit. The weights  $w_f$  and  $w_v$  set the relative importance of frequency and voltage objectives in the optimisation.

Subject to:

$$f_{\min} \leq f(t) \leq f_{\max}, \quad \left| \frac{df}{dt} \right| \leq \text{RoCoF}_{\max}, \quad V_{\min} \leq V(t) \leq V_{\max}, \quad 0 \leq x \leq 1, \quad (9)$$

where  $f_{\min}$  and  $f_{\max}$  denote allowable frequency bounds,  $\text{RoCoF}_{\max}$  is the maximum permissible rate-of-change of frequency, and  $V_{\min}$ ,  $V_{\max}$  are the voltage limits as per grid code requirements.

#### 2.2.3. Optimization algorithm

The GFM:GFL penetration ratio is optimised using Particle Swarm Optimisation (PSO), where each particle represents a candidate share  $x \in [0, 1]$  and the objective is to maximise the composite stability index  $J(x)$ . The particle update rule is:

$$\begin{aligned} \Delta x_i^{k+1} &= \omega \Delta x_i^k + c_1 r_1 (x_{i,\text{best}} - x_i^k) + c_2 r_2 (x_{\text{global}} - x_i^k); \\ x_i^{k+1} &= x_i^k + \Delta x_i^{k+1} \end{aligned} \quad (10)$$

where  $\Delta x_i$  is the GFM-share update,  $\omega$  the inertia weight,  $c_1, c_2$  the acceleration gains, and  $r_1, r_2$  random terms. Here,  $x_{i,\text{best}}$  and  $x_{\text{global}}$  are the particle and global best values. Iterations continue until the GFM share that maximises  $J(x)$  is found.

### 2.3. Comparative evaluation of Droop and VSM control

Following the parametric analysis in contribution 2, this contribution compares *Droop* and *VSM* controls at 50 % and 75 % GFM shares, evaluating frequency-voltage performance and black-start capability. Droop and VSM serve as alternative GFM outer loops: for each GFM share  $x$ , BESS units operate either Droop or VSM, with the remainder in GFL mode; no inverter uses both controls simultaneously.

#### 2.3.1. Droop control

Droop control applies proportional  $P$ - $f$  and  $Q$ - $V$  laws:

$$\Delta f = -k_p (P - P^*), \quad \Delta V = -k_q (Q - Q^*), \quad (11)$$

where  $k_p$  and  $k_q$  are droop coefficients. It enables simple power sharing but lacks inertia emulation, making weak grids more sensitive to frequency disturbances. This algebraic structure enables a rapid response to disturbances, a behaviour also reported in prior studies, such as Tozak et al. [12] and the experimental results of Nassif et al. [15], where droop-based GFM inverters consistently exhibited the quickest frequency arrest.

#### 2.3.2. VSM control

VSM incorporates swing dynamics to emulate synchronous machine behaviour:

$$M \frac{d\omega}{dt} + D(\omega - \omega^*) = P^* - P, \quad (12)$$

where  $M$  is virtual inertia and  $D$  is damping. This provides an inertia-like response and enhanced damping, improving transient stability.

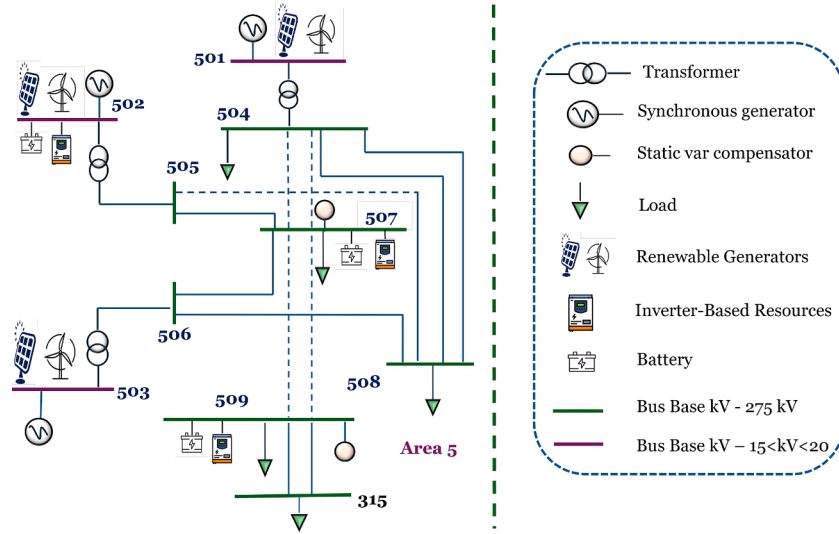


Fig. 3. Simplified Australian power system test network (Area 5).

These additional dynamic elements slow the initial response but significantly enhance oscillation attenuation and settling behaviour. This inertia damping effect is discussed in the VSM literature [12], where VSM implementations demonstrate smoother trajectories and superior transient damping compared with droop control.

### 2.3.3. Black start capability

In weak-grid conditions, droop control can cause frequency drift while VSM offers smoother recovery, though their effectiveness depends on system conditions [26]. Here,  $P$  and  $Q$  denote active and reactive power, respectively, and  $\sigma_G(t)$  represents the generator state. This can be expressed by droop-style balance:

$$\sum_j (P_{ij} + jQ_{ij}) = (P_L + jQ_L) - (P_G + jQ_G), \quad (13)$$

and VSM-like cranking dynamics:

$$\begin{aligned} P_G(t) &= \sigma_G(t) P_{\text{avail}} - (1 - \sigma_G(t)) P_{\text{crank}}, \\ Q_G(t) &= \sigma_G(t) Q_{\text{avail}} - (1 - \sigma_G(t)) Q_{\text{crank}}, \end{aligned} \quad (14)$$

which capture system balancing and generator startup in restoration. The next section introduces the network and cases for evaluating the proposed methodology.

## 3. Network and cases considered

This section outlines the network setup and case studies used to demonstrate the validity of the proposed methodology.

### 3.1. Network considered

The test system is based on Area 5 of the simplified Australian 14-bus generator network (Fig. 3), which has been widely used in stability studies of inverter-dominated grids. Detailed specifications are provided in Gibbard and Vowles [27], Al-Masood [28]. Area 5 represents a weak-grid environment with synchronous machines, inverter-based renewables, and interconnected loads, making it well-suited for evaluating the proposed coordinated BESS control framework. In this work, the bus hosting both GFM and GFL units represents the medium-voltage collector bus of a multi-inverter BESS plant, where several inverter blocks operate in parallel [29,30]. In line with modern practice, some blocks are configured in GFM mode to establish local voltage and frequency, while others remain GFL for power dispatch and charging control [16,21].

### 3.2. Cases considered

The simulation cases correspond directly to the three methodological contributions. Contribution 1 considers four benchmarks: a baseline without IBRs, GFL-only, GFM-only, and coordinated GFM–GFL control governed by the severity index, thereby illustrating the incremental benefits of adaptive coordination. Contribution 2 extends this through a systematic parametric sweep of GFM shares  $x \in \{0, 0.25, 0.5, 0.75, 1.0\}$ , assessed using a composite frequency–voltage stability index that consolidates nadir, overshoot, settling behavior, and voltage margins into a unified measure of dynamic resilience. Contribution 3 compares Droop and VSM control at representative shares of 50 % and 75 %, focusing not only on transient stability but also on black start capability, which provides insight into post-disturbance recovery. All simulations are conducted in Power System Simulation for Engineering (PSS®E 36.2) with Python 3.12 on a 12<sup>th</sup> Generation Intel® Core™ i7-1265U Central Processing Unit (CPU) with a base clock speed of 1.80 gigahertz (GHz) and 32 gigabytes (GB) of Random Access Memory (RAM); which is appropriate for system-level assessment of transmission networks; experimental and HIL validation are left for future work. The next section discusses the corresponding results in detail.

## 4. Results and discussion

This section validates the methodology through dynamic simulations, presenting results from coordination analysis to parametric evaluation and strategy comparison.

### 4.1. Severity-aware GFM–GFL coordination

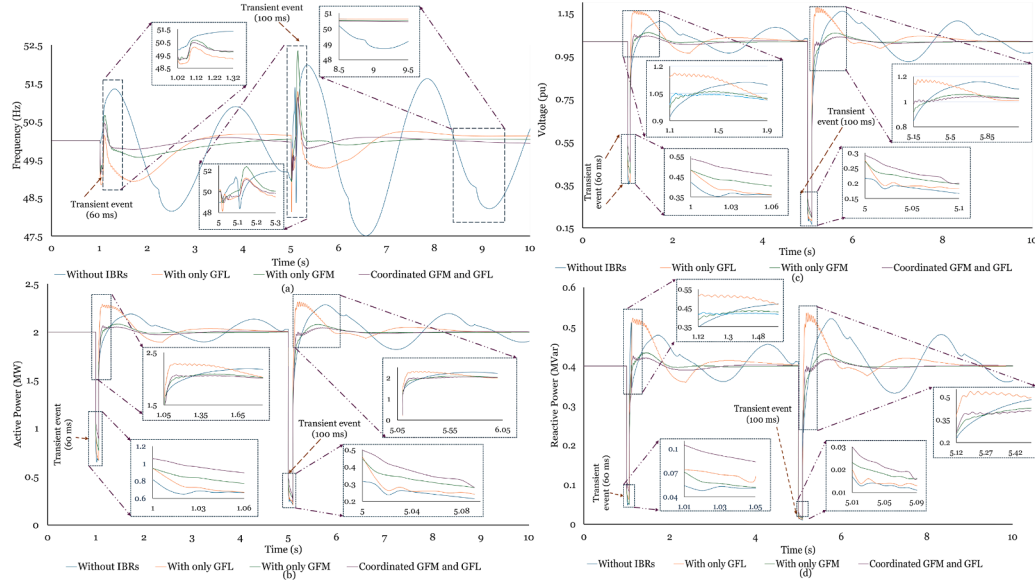
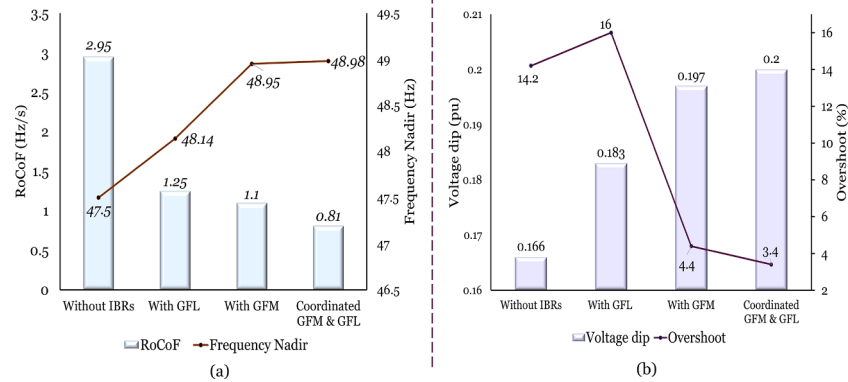
The proposed severity-aware GFM–GFL coordination strategy is benchmarked against three baselines: (i) no IBRs, (ii) GFL-only operation, and (iii) GFM-only operation. Quantitative results are summarised in Table 3, with the corresponding dynamic responses shown in Figs. 4 and 5.

Without IBRs, the system exhibits the most critical instability, with a frequency nadir of 47.5 Hz, 14.2 % overshoot, RoCoF of 2.95 Hz/s, and the highest severity index ( $S = 4.77$ ). Under GFL-only operation, the nadir improves to 48.1 Hz and RoCoF falls to 1.25 Hz/s, but overshoot increases to 16 %, highlighting the limited ability of PLL-driven current control to stabilise post-fault voltage. In contrast, GFM-only operation reduces overshoot to 4.4 % and raises the nadir to 48.95 Hz, yet voltage recovery is slower (minimum 0.197 p.u. at 5.1 s), indicating degraded performance under prolonged low-voltage conditions.

**Table 3**

Voltage and frequency performance across four cases.

| Case                  | Peak vol- tage (p.u.) | Peak vol- tage time (s) | Over- shoot (%) | shoot | Lowest vol- tage (p.u.) | Lowest vol- tage time (s) | Ave- rage Ro- CoF (Hz/s) | Frequ- ency Nadir (Hz) | Severity Index (S) |
|-----------------------|-----------------------|-------------------------|-----------------|-------|-------------------------|---------------------------|--------------------------|------------------------|--------------------|
| Without IBRs          | 1.159                 | 5.75                    | 14.2            |       | 0.166                   | 5.10                      | 2.9544                   | 47.5071                | 4.77               |
| With GFL              | 1.177                 | 5.23                    | 16.0            |       | 0.183                   | 5.09                      | 1.2470                   | 48.1359                | 4.10               |
| With GFM              | 1.061                 | 1.46                    | 4.4             |       | 0.197                   | 5.10                      | 1.0950                   | 48.9474                | 3.50               |
| Coordinated GFM & GFL | 1.051                 | 1.15                    | 3.4             |       | 0.200                   | 5.10                      | 0.8081                   | 48.9768                | 3.44               |

**Fig. 4.** Dynamic response at Bus 509 across four cases: (a) frequency, (b) active power, (c) voltage, and (d) reactive power.**Fig. 5.** Comparative performance across four control schemes: (a) RoCoF and frequency nadir, and (b) voltage dip and overshoot.

The coordinated GFM–GFL strategy delivers the most balanced response: overshoot is limited to 3.4 %, RoCoF is reduced to 0.81 Hz/s (a 73 % improvement relative to the no-IBR case), and the nadir increases to 48.98 Hz, yielding the lowest severity index ( $S = 3.44$ ). These gains arise from the complementary roles of GFL and GFM modes: GFL control provides fast, current-limited active–reactive support, while GFM control supplies robust voltage formation and inertia-like frequency support as severity increases. Through the severity-dependent blending factor  $\lambda(t)$ , the controller smoothly shifts from GFL-dominant behaviour during mild events to greater GFM contribution during deeper faults.

From a control viewpoint, the shared bus functions as a multi-inverter BESS plant where GFM units act as stiff voltage sources and GFL units operate as PLL-synchronised current sources. Guided by the severity index, GFM provides a strong local voltage–frequency reference while GFL delivers rapid current support, and shifting part of the plant from

pure GFL to blended GFM–GFL operation improves frequency nadir, reduces overshoot, and accelerates voltage recovery (Table 3)

Overall, the results show that the severity-aware coordinated strategy consistently outperforms either single-mode configuration, combining their strengths to enhance stability, damping, and resilience under grid-code fault conditions, thereby substantiating the claims of Contribution 1.

#### 4.2. Optimisation of GFM:GFL penetration ratios

The second contribution examines the impact of varying GFM:GFL penetration ratios ( $x \in \{0, 0.25, 0.5, 0.75, 1.0\}$ ) while maintaining a fixed total generation of 330 MW. Frequency and voltage dynamics are evaluated at bus 502, with results summarised in Tables 4–5 and illustrated in

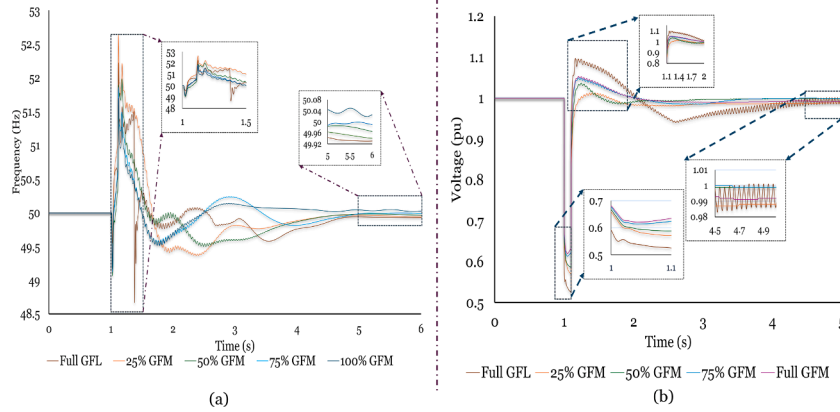


Fig. 6. Frequency (a) and voltage (b) responses at Bus 502 under varying GFM/GFL penetration levels.

**Table 4**  
Frequency stability metrics under different GFM:GFL ratios.

| $x$  | Nadir (Hz) | Peak (Hz) | $\Delta f$ (Hz) | Settling Time (s) | IAE   | ITAE  |
|------|------------|-----------|-----------------|-------------------|-------|-------|
| 0.00 | 48.71      | 52.21     | 3.50            | 5.05              | 1.361 | 2.364 |
| 0.25 | 49.22      | 52.61     | 3.39            | 4.90              | 0.940 | 1.750 |
| 0.50 | 49.09      | 51.98     | 2.89            | 4.65              | 0.815 | 1.509 |
| 0.75 | 49.15      | 51.84     | 2.69            | 4.50              | 0.661 | 1.124 |
| 1.00 | 49.19      | 51.69     | 2.50            | 4.21              | 0.568 | 0.937 |

**Table 5**  
Voltage stability metrics under different GFM:GFL ratios.

| $x$  | $V_{\min}$ (p.u.) | $V_{\text{over}}$ (p.u.) | $\Delta V$ (p.u.) | Objective Function $J$ |
|------|-------------------|--------------------------|-------------------|------------------------|
| 0.00 | 0.525             | 1.10                     | 0.525             | 0.175                  |
| 0.25 | 0.572             | 1.02                     | 0.448             | 0.456                  |
| 0.50 | 0.588             | 1.03                     | 0.442             | 0.497                  |
| 0.75 | 0.614             | 1.04                     | 0.426             | 0.569                  |
| 1.00 | 0.621             | 1.04                     | 0.419             | 0.619                  |

Figs. 6 and 7. The analysis demonstrates a consistent trend: higher GFM penetration yields progressively enhanced stability across both domains.

At full GFL operation ( $x = 0$ ), frequency resilience is weakest, with a nadir of 48.71 Hz, overshoot of 3.50 Hz, settling time of 5.05 s, and the lowest  $V_{\min}$  of 0.525 p.u. Introducing partial GFM support ( $x = 0.25$ ) raises the nadir and shortens settling time, but error indices remain high (IAE = 0.940, ITAE = 1.750). A more balanced response appears at  $x = 0.5$ , where overshoot falls to 2.89 Hz, settling time reduces to 4.65 s, and both IAE and ITAE drop markedly. The strongest performance occurs at full GFM ( $x = 1.0$ ), with the highest nadir (49.19 Hz), lowest overshoot (2.50 Hz), fastest settling (4.21 s), and minimum error values (IAE = 0.568, ITAE = 0.937). Voltage stability also improves monotonically, as  $V_{\min}$  increases from 0.525 to 0.621 p.u. and  $\Delta V$  contracts from 0.525 to 0.419 p.u.

This monotonic trend reflects the complementary control roles of GFM and GFL inverters. For a fixed 330 MW BESS, increasing the GFM share  $x$  shifts more capacity from current-following to voltage-forming, enhancing virtual inertia and voltage stiffness at bus 502 and thus improving nadir, reducing overshoot, and accelerating damping. The remaining GFL units exploit this strengthened reference to deliver rapid active and reactive current injection. Together, this coordination yields an  $\approx 60\%$  reduction in ITAE, an  $\approx 20\%$  narrowing of  $\Delta V$ , and a 3.5 fold increase in  $J(x)$  between  $x = 0$  and  $x = 1$ .

$J(x)$  rises steadily with GFM share, peaking at 0.619 when  $x = 1$ . Higher GFM fractions deliver stability levels close to full GFM, improving frequency nadir, damping, recovery, and voltage support. This confirms Contribution 2 and validates the optimisation framework in weak-grid conditions.

**Table 6**  
Comparison of Droop and VSM control at 50 % and 75 % GFM share.

| Metric                   | Droop (50 %) | Droop (75 %) | VSM (50 %) | VSM (75 %) |
|--------------------------|--------------|--------------|------------|------------|
| Frequency nadir (Hz)     | 49.09        | 49.15        | 47.9       | 49.07      |
| Peak overshoot (Hz)      | 51.91        | 51.84        | 50.95      | 51.77      |
| Settling frequency (Hz)  | 49.91        | 48.96        | 49.98      | 49.62      |
| Voltage dip (p.u.)       | 0.57         | 0.61         | 0.63       | 0.62       |
| Voltage overshoot (p.u.) | 1.02         | 1.04         | 1.05       | 1.04       |

## 5. Comparative evaluation of Droop and VSM control

The final contribution compares Droop and VSM performance at 50 % and 75 % GFM (Table 6, Fig. 8), highlighting trade-offs in transient response and robustness. Four cases are evaluated: Droop–50 %, Droop–75 %, VSM–50 %, VSM–75 % with each scenario using a single control law.

Droop control provides faster and more bounded transients, with nadirs of 49.09 Hz (50 % GFM) and 49.15 Hz (75 % GFM) and overshoots of 51.91–51.84 Hz. However, its proportional  $P$ – $f$  and  $Q$ – $V$  response lacks inertia emulation, leading to reduced long-term frequency restoration, as reflected in the lower settling frequency at 75 % GFM (48.96 Hz). VSM control, by contrast, delivers superior inertial buffering and damping, improving nadir recovery and raising settling frequency to 49.98 Hz (50 % GFM) and 49.62 Hz (75 % GFM), albeit with slightly higher voltage overshoot (1.05 p.u. versus 1.02 p.u. for Droop).

Sensitivity analysis (Fig. 9) shows that varying the droop gain  $k_p$  (0.1–0.5) or the VSM inertia constant  $M$  (1–5) produces only marginal changes in voltage dip ( $\approx 0.565$ – $0.570$  p.u.) and overshoot ( $\approx 1.01$ – $1.05\%$ ). This narrow response band confirms that both control methods are highly robust to parameter tuning.

Overall, Droop is favourable when rapid transient response and bounded overshoot are prioritised, whereas VSM offers improved frequency recovery and resilience in low-inertia weak grids. The complementary strengths identified here reinforce the need for coordinated GFM–GFL deployment in high-renewable networks.

### 5.1. Black start capability

The black start analysis was conducted with 75 % GFM share for both Droop and VSM strategies, to verify whether coordinated GFM–GFL control can reliably achieve restoration rather than emphasising differences between the two modes. Results (Table 7, Fig. 10) confirm that both strategies enable successful staged re-energisation of the system through GFM–BESS resources, including load pickup, generator cranking, and island synchronisation.

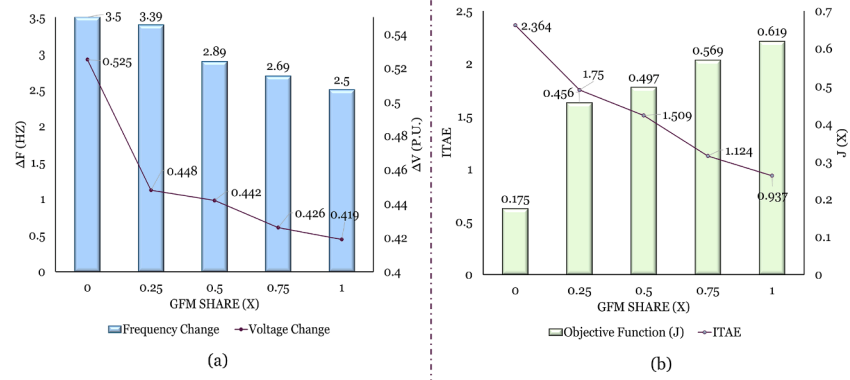


Fig. 7. Variation of (a) frequency and voltage indicators, and (b) objective function (J) and ITAE across different GFM penetration levels.

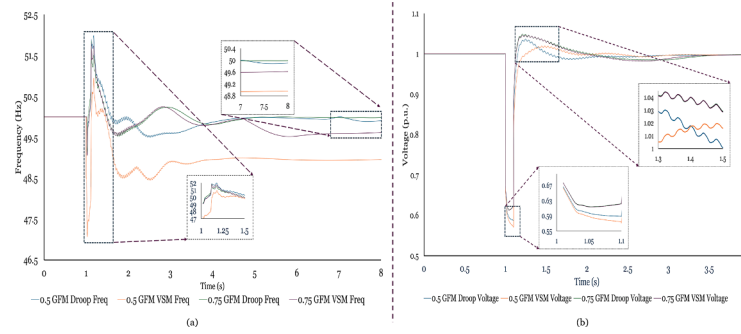


Fig. 8. Frequency (a) and voltage (b) responses of Droop and VSM controls at 50% and 75% GFM penetration levels.

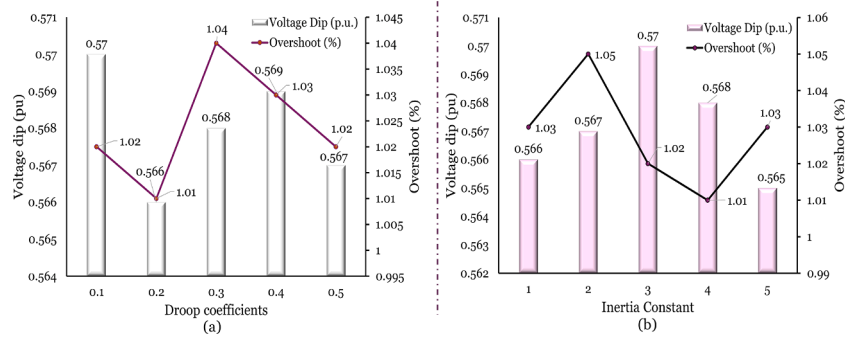


Fig. 9. Variation of (a) droop coefficients and (b) inertia constants.

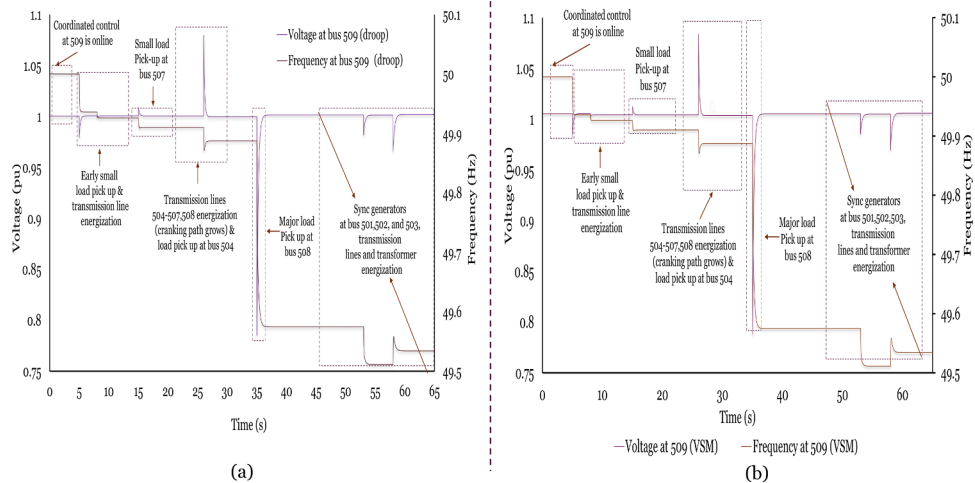


Fig. 10. Black start restoration at Bus 509 using GFM-BESS showing voltage and frequency performance with (a) Droop control and (b) VSM control.

**Table 7**  
Black start restoration sequence enabled by GFM-BESS.

| Time (s) | Action                                                       |
|----------|--------------------------------------------------------------|
| 0        | Energise Bus 509 in GFM mode (black start resource)          |
| 5        | Staged load pickup at Bus 509                                |
| 8        | Staged load pickup at Bus 315                                |
| 12       | Close line 507–509: energise cranking path                   |
| 15       | Load pickup at Bus 507                                       |
| 20       | Generator cranking at Bus 507                                |
| 22       | Continue energisation toward backbone                        |
| 26       | Additional load pickup at Bus 507                            |
| 30       | Close line 504–508: transmission strengthening               |
| 35       | Load pickup at Bus 508                                       |
| 40       | Transformer energisation (Bus 501–504)                       |
| 45       | Generator cranking at Bus 501                                |
| 50       | Backbone energisation (504–508 and 505–507)                  |
| 50–60    | Synchronisation of islands/generators at Buses 501, 502, 503 |

Droop and VSM show broadly similar restoration performance, with Droop offering faster initial transients and VSM providing smoother trajectories with reduced oscillations. This close alignment demonstrates that the coordinated framework itself is capable of facilitating black start restoration in weak grids when sufficient GFM share is present. These findings highlight the viability of the proposed control architecture, which not only enhances dynamic stability under disturbances but also enables full grid restoration. This concludes the evaluation of the three contributions and leads to the final discussion on conclusions and future work.

## 6. Conclusion and future work

This paper has addressed the stability challenges of renewable-dominated weak grids by proposing a severity-aware BESS control framework where GFL and GFM inverters operate in a blended manner governed by a severity index. The framework ensures efficient current-following during mild disturbances while adaptively transitioning to voltage-forming dominance under severe faults, thus combining operational efficiency with resilience. Validation through parametric analysis confirmed monotonic improvements in frequency nadir, RoCoF suppression, and voltage stability with increasing GFM share, while comparative evaluation of Droop and VSM strategies highlighted the trade-off between Droop's faster transients and VSM's superior damping and black start capability.

The findings highlight severity-aware GFM-GFL coordination as a practical pathway to resilient inverter-dominated grids, directly supporting SDG 7. Nonetheless, the study is limited to simulation-based validation on simplified test systems. The detailed RMS-domain simulations presented are suitable for system-level evaluation under weak grid conditions. By contrast, hardware or hardware-in-the-loop (HIL) implementation would require extensive laboratory infrastructure and is therefore identified as an important direction for future work. Future research should extend the framework to larger networks with renewable variability, demand fluctuations, and real-world contingencies, while exploring real-time optimisation of GFM:GFL allocation, hybrid Droop-VSM schemes, advanced black start protocols, and experimental or HIL implementations (e.g., on RTDS/OPAL-RT platforms with physical inverters) to bridge the gap between simulations and field deployment.

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## Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

## CRediT authorship contribution statement

**Sadnan Sakib:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Conceptualization; **M. J. Hossain:** Writing – review & editing, Validation, Supervision, Software, Methodology, Investigation, Conceptualization.

## Declaration of competing interest

The authors declare no conflicts of interest.

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